MILITARY SPECIFICATION SHEET

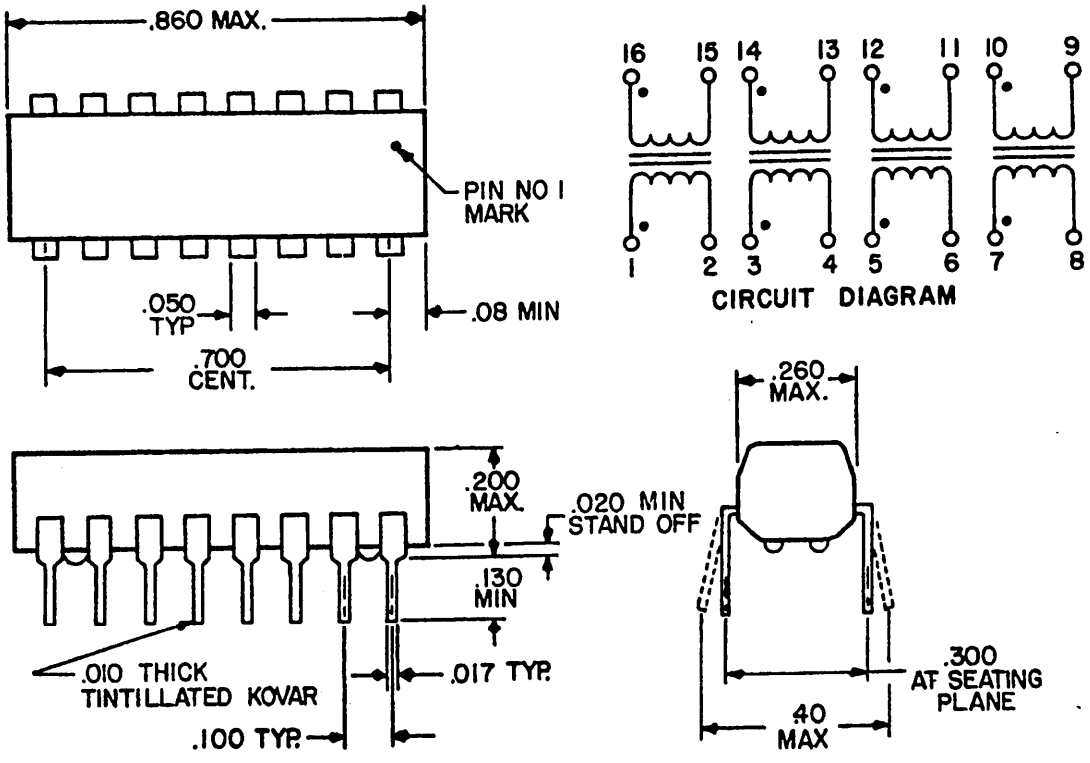
TRANSFORMERS, PULSE, LOW POWER,

TYPE TP7SX2100MZ

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-21038.

Ⓐ This specification is inactive for new design after the date of this specification.




| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .010 | .25 | .200 | 5.08 |
| .017 | .43 | .260 | 6.60 |
| .020 | .51 | .300 | 7.62 |
| .050 | 1.27 | .40 | 10.16 |
| .08 | 2.03 | .700 | 17.78 |
| .100 | 2.54 | .860 | 21.84 |
| .130 | 3.30 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).

FIGURE 1. Dimensions and configuration.

Ⓐ denotes changes



PSC 5950

Source: https://assist.dla.mil -- Downloaded: 20261O-19T08:33Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics.

| Dash no. | Primary inductance (±20%) | Primary ET constant (min) | Interwinding capacitance (max) | Leakage inductance (max) | Primary DCR (max) | Secondary DCR (max) |
|---|---|---|---|---|---|---|
| | µH | Vµs | pF | µH | ohms | ohms |
| 001 | 8 | 1.0 | 5 | .50 | .16 | .12 |
| 002 | 30 | 2.1 | 7 | .50 | .19 | .13 |
| 003 | 57 | 2.8 | 7 | .50 | .34 | .25 |
| 004 | 130 | 2.0 | 7 | .50 | .34 | .25 |
| 005 | 230 | 2.7 | 9 | .50 | .55 | .32 |
| 006 | 430 | 3.8 | 11 | .50 | .75 | .38 |

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Turns ratio: 2:1.

Duty cycle: Continuous.

Average power rating: 250 mW.

Case: Encapsulated.

Terminals: Pin.

Terminal material: Type K per MIL-STD-1276.

Weight: 1 gram, maximum.

Terminal strength: Method 2004, MIL-STD-883, test condition B2.

Dielectric withstanding voltage:
At atmospheric pressure: Method 301, MIL-STD-202, 100 volts rms for 60 seconds between windings and ground.
Barometric pressure (reduced): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. Not less than 1,000 megohms.

Electrical characteristics: See table I.
Voltage-time product (ET constant): The voltage-time product shall be determined by multiplying the amplitude of the applied pulse by the time required for the magnetizing current to deviate 20 percent from the value of the linearly extrapolated ramp portion of the magnetizing-current waveform in the region of core saturation.

Life (expectancy "X"): Applicable.



Source: https://assist.dla.mil -- Downloaded: 20261O-19T08:33Z
Check the source to verify that this is the current version before use.

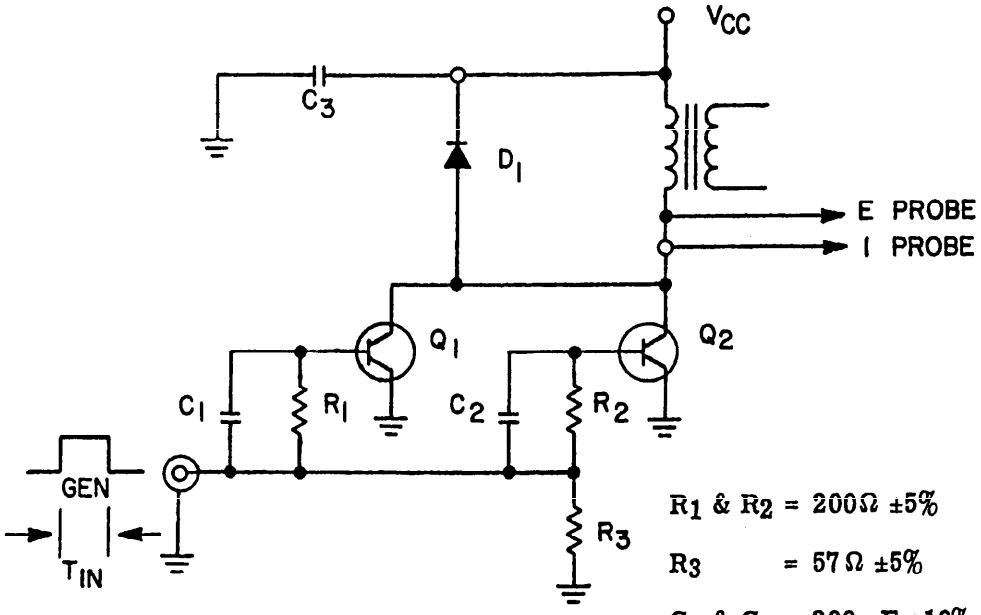
Ambient temperature: 90°C.

Temperature rise: 40°C (250 mW).

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock (specified pulse): Method 213, test condition H, MIL-STD-202.

Part number: M21038/17 -(dash number from table I).




$R_1$ & $R_2$ = 200Ω ±5%

$R_3$ = 57Ω ±5%

$C_1$ & $C_2$ = 200 pF ±10%

$C_3$ = 15 to 25 μF @ 25 Vdc

$Q_1$ & $Q_2$ = 2N3725 NPN Transistors

$D_1$ = 1N4531 Diode

FIGURE 2. Test circuit.

Ⓐ Quality assurance provisions:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Groups A and B tests of MIL-T-21038 shall be applicable.

Part number: M21038/17- (dash number from table I).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:33Z
Check the source to verify that this is the current version before use.

Preparing activity:
Air Force - 85

(Project 5950-0587-09)



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:33Z
Check the source to verify that this is the current version before use.